

Commonly Used Distribution – II

Yi-Fen Liu

Department of IECS, FCU

References:

- Dimitri P. Bertsekas and John N. Tsitsiklis, *Introduction to Probability*, Chapter 3
- Walpole R. E., Myers R. H., Myers S. L. and Ye K., *Probability & Statistics for Engineers & Scientists*, Ch. 5
- William Navidi, *Statistics for Engineers and Scientists*, Chapter 4
- Slides are credited from Prof. Berlin Chen, NTNU.

CONTINUOUS DISTRIBUTION

The Normal Distribution

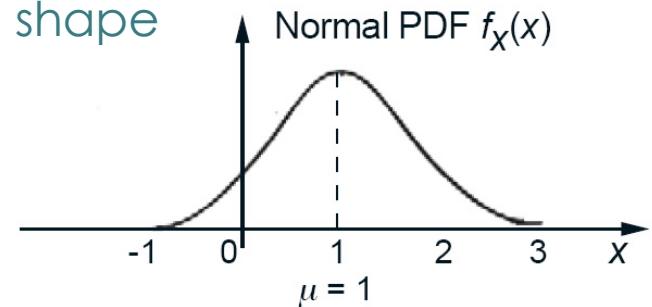
- The **normal distribution** (also called the Gaussian distribution) is by far the most commonly used distribution in statistics. This distribution provides a good model for many, although not all, continuous populations
- The normal distribution is continuous rather than discrete. The mean of a normal population may have any value, and the variance may have any positive value

pdf, Mean and Variance of Normal

- The probability density function of a normal population with mean μ and variance σ^2 is given by

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty \leq x \leq \infty$$

bell shape

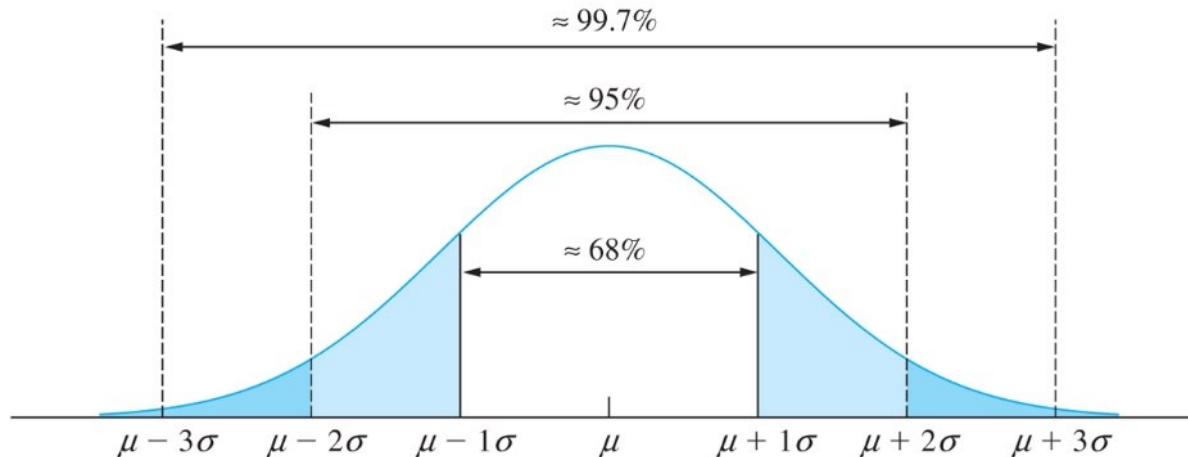


- If $X \sim N(\mu, \sigma^2)$, then the mean and variance of X are given by

$$\mu_X = \mu$$

$$\sigma_X^2 = \sigma^2$$

68-95-99.7% Rule



- The above figure represents a plot of the normal probability density function with mean μ and standard deviation σ . Note that the curve is symmetric about μ , so that μ is the median as well as the mean. It is also the case for the normal population
 - About 68% of the population is in the interval $\mu \pm \sigma$
 - About 95% of the population is in the interval $\mu \pm 2\sigma$
 - About 99.7% of the population is in the interval $\mu \pm 3\sigma$

Standard Units

- The proportion of a normal population that is within a given number of standard deviations of the mean is the same for any normal population
- For this reason, when dealing with normal populations, we often convert from the units in which the population items were originally measured to **standard units**
- Standard units tell how many standard deviations an observation is from the population mean

Standard Normal Distribution

- In general, we convert to **standard units** by subtracting the mean and dividing by the standard deviation. Thus, if x is an item sampled from a normal population with mean μ and variance σ^2 , the standard unit equivalent of x is the number z , where

$$z = (x - \mu) / \sigma$$

- The number z is sometimes called the “z-score” of x . The z-score is an item sampled from a normal population with mean 0 and standard deviation of 1. This normal distribution is called the **standard normal distribution**

– pdf $f_z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}, \quad -\infty \leq z \leq \infty$

The Proof

- Mean and Variance of the **Normal** Random Variable X

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty \leq x \leq \infty$$

? (see the end of Sec. 4.5)

$$\text{Let } Y = \frac{X - \mu}{\sigma} \Rightarrow f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}}, \quad -\infty \leq y \leq \infty$$

$$\mathbf{E}[Y] = \int_{-\infty}^{\infty} y \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy = -\frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} \Big|_{-\infty}^{\infty} = 0$$

$$\Rightarrow \mathbf{E}[X] = \sigma \mathbf{E}[Y] + \mu = 0 + \mu = \mu$$

$$\text{var}[Y] = \int_{-\infty}^{\infty} (y - \mathbf{E}[Y])^2 \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y^2 e^{-\frac{y^2}{2}} dy = \left[\frac{1}{\sqrt{2\pi}} \cdot -ye^{-\frac{y^2}{2}} \Big|_{-\infty}^{\infty} \right] + \left[\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy \right]$$

$$= 0 + 1$$

$$= 1 \quad \because \text{var}(Y) = \frac{\text{var}(X)}{\sigma^2} \therefore \text{var}(X) = \sigma^2$$

$$\left(\because \frac{d\left(-ye^{-\frac{y^2}{2}}\right)}{dy} = y^2 e^{-\frac{y^2}{2}} - e^{-\frac{y^2}{2}} \right)$$

1

Examples

1. Q: Aluminum sheets used to make beverage cans have thicknesses (in thousandths of an inch) that are normally distributed with mean 10 and standard deviation 1.3. A particular sheet is 10.8 thousandths of an inch thick. Find the z-score:

$$\text{Ans.: } z = (10.8 - 10)/1.3 = 0.62$$

2. Q: Use the same information as in 1. The thickness of a certain sheet has a z-score of -1.7. Find the thickness of the sheet in the original units of thousandths of inches:

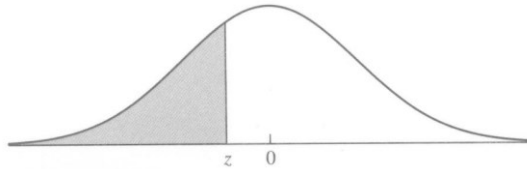
$$\text{Ans.: } -1.7 = (x - 10)/1.3 \quad x = -1.7(1.3) + 10 = 7.8$$

Finding Areas Under the Normal Curve

- The proportion of a normal population that lies within a given interval is equal to the area under the normal probability density above that interval. This would suggest integrating the normal pdf; this integral have **no closed form solution**
- So, the areas under the curve are approximated numerically and are available in **Table A.2** (Z-table). This table provides area under the curve for the standard normal density. We can convert any normal into a standard normal so that we can compute areas under the curve
 - The table gives the area in the **left-hand tail** of the curve
 - Other areas can be calculated by subtraction or by using the fact that the total area under the curve is 1

Z-Table (1/2)

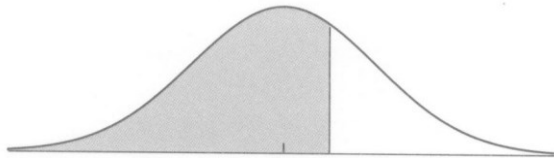
TABLE A.2 Cumulative normal distribution (z table)



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.6	.0002	.0002	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001
-3.5	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

Z-Table (2/2)

TABLE A.2 Cumulative normal distribution (continued)

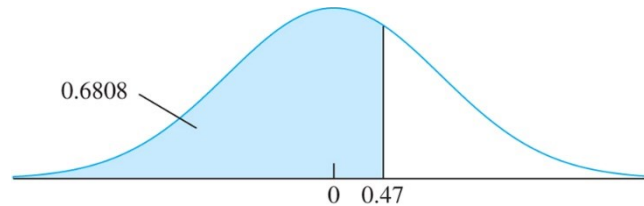


z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998
3.5	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998
3.6	.9998	.9998	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999

Examples

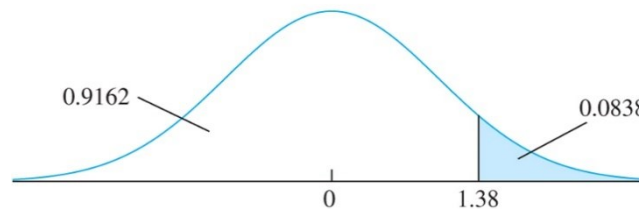
1. Q: Find the area under normal curve to the left of $z = 0.47$

Ans.: From the z table, the area is 0.6808



2. Q: Find the area under the curve to the right of $z = 1.38$

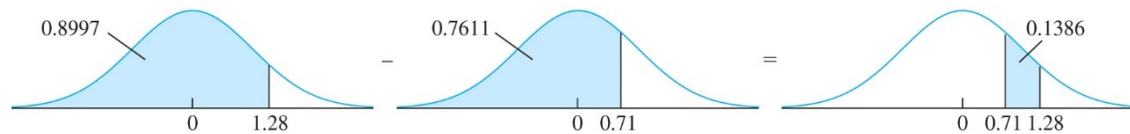
Ans.: From the z table, the area to the left of 1.38 is 0.9162.
Therefore the area to the right is $1 - 0.9162 = 0.0838$



More Examples

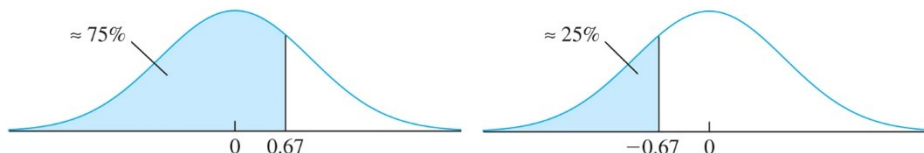
1. Q: Find the area under normal curve between $z = 0.71$ and $z = 1.28$

Ans.: The area to the left of $z = 1.28$ is 0.8997. The area to the left of $z = 0.71$ is 0.7611. So the area between is $0.8997 - 0.7611 = 0.1386$



2. Q: What z-score corresponds to the 75th percentile of a normal curve? The 25th percentile?

Ans.: To answer this question, we use the z table in reverse. We need to find the z-score for which 75% of the area of curve is to the left. From the body of the table, the closest area to 75% is 0.7486, corresponding to a z-score of 0.67



$$\Phi(-z) = 1 - \Phi(z),$$

for all z

Estimating the Parameters of Normal

- $X_1, \dots, X_n$ are a random sample from a $N(\mu, \sigma^2)$ distribution, μ is estimated with the sample mean $\bar{X}$ and σ^2 is estimated with the sample variance s^2

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

unbiased estimator

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

asymptotically unbiased estimator?

- As with any sample mean, the uncertainty in $\bar{X}$ is $\sigma / \sqrt{n}$ which we replace with $s / \sqrt{n}$, if σ is unknown. The mean is an unbiased estimator of μ

Sample Variance is an Asymptotically Unbiased Estimator (1/3)

- Sample variance s^2 is an asymptotically unbiased estimator of the population variance σ^2

$$\begin{aligned}\mathbf{E}[s^2] &= \mathbf{E}\left[\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2\right] \\&= \mathbf{E}\left[\frac{1}{n} \sum_{i=1}^n (X_i^2 - \underline{2X_i \cdot \bar{X}} + \bar{X}^2)\right] \\&= \mathbf{E}\left[\frac{\left(\sum_{i=1}^n X_i^2\right) - \underline{2n \cdot \bar{X}^2} + n\bar{X}^2}{n}\right] \\&= \mathbf{E}\left[\frac{\left(\sum_{i=1}^n X_i^2\right) - n\bar{X}^2}{n}\right] \\&= \frac{\left(\sum_{i=1}^n \mathbf{E}[X_i^2]\right) - n \cdot \mathbf{E}[\bar{X}^2]}{n}\end{aligned}$$

$$\begin{aligned}s^2 &= \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \\&\Rightarrow \sum_{i=1}^n X_i = n \cdot \bar{X}\end{aligned}$$

Sample Variance is an Asymptotically Unbiased Estimator (2/3)

$$\begin{aligned}\text{Var}(\bar{X}) &= \frac{\sigma^2}{n} = \mathbf{E}[\bar{X}^2] - (\mathbf{E}[\bar{X}])^2 \\ \Rightarrow \mathbf{E}[\bar{X}^2] &= \frac{\sigma^2}{n} + (\mathbf{E}[\bar{X}])^2 = \frac{\sigma^2}{n} + \mu^2\end{aligned}$$

$$\begin{aligned}\mathbf{E}[s^2] &= \frac{\left(\sum_{i=1}^n \mathbf{E}[X_i^2] \right) - n \cdot \mathbf{E}[\bar{X}^2]}{n} \\ &= \frac{n(\sigma^2 + \mu^2) - n\left(\frac{\sigma^2}{n} + \mu^2\right)}{n} \\ &= \frac{n-1}{n} \sigma^2 \quad \text{when } n = \infty \rightarrow \sigma^2\end{aligned}$$

$$\begin{aligned}\text{Var}(X_i) &= \sigma^2 = \mathbf{E}[X_i^2] - (\mathbf{E}[X_i])^2 \\ \Rightarrow \mathbf{E}[X_i^2] &= \sigma^2 + (\mathbf{E}[X_i])^2 = \sigma^2 + \mu^2\end{aligned}$$

The size of the observed sample

Sample Variance is an Asymptotically Unbiased Estimator (3/3)

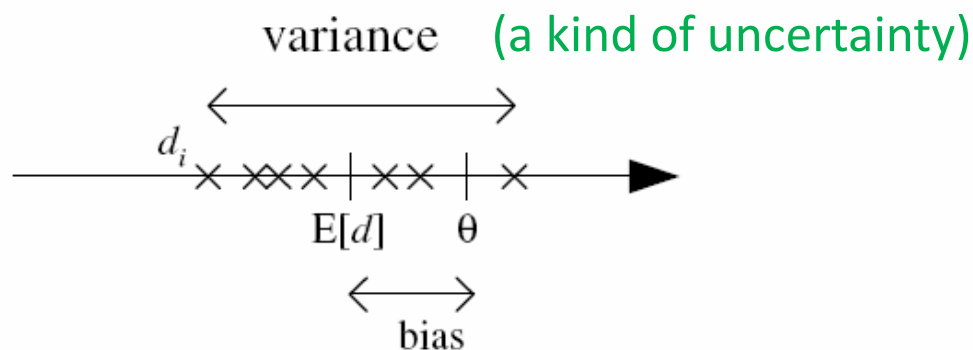
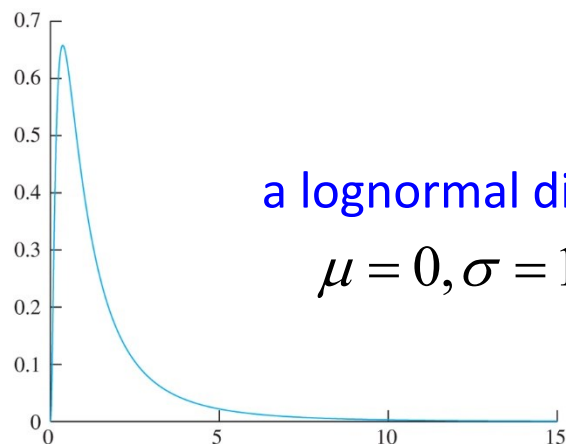


Figure 4.1: θ is the parameter to be estimated. d_i are several estimates (denoted by 'x') over different samples. Bias is the difference between the expected value of d and θ . Variance is how much d_i are scattered around the expected value. We would like both to be small.

The Lognormal Distribution

- For data that contain outliers (on the right of the axis), the normal distribution is generally not appropriate. The **lognormal distribution**, which is related to the normal distribution, is often a good choice for these data sets
- If $X \sim N(\mu, \sigma^2)$, then the random variable $Y = e^X$ has the lognormal distribution with parameters μ and σ^2
- If Y has the lognormal distribution with parameters μ and σ^2 , then the random variable $X = \ln Y$ has the $N(\mu, \sigma^2)$ distribution

Probability Density Function



a lognormal distribution with parameters

$$\mu = 0, \sigma = 1$$

pdf, Mean and Variance of Lognormal

- The pdf of a lognormal random variable with parameters μ and σ^2 is

$$f_Y(y) = \begin{cases} \frac{1}{\sigma y \sqrt{2\pi}} \exp\left[-\frac{1}{2\sigma^2} (\ln y - \mu)^2\right], & y > 0 \\ 0, & \text{otherwise} \end{cases}$$

- The mean $\mathbf{E}[Y]$ and variance $\text{Var}(Y)$ are given by

$$\mathbf{E}[Y] = e^{\mu + \sigma^2/2} \qquad \text{Var}(Y) = e^{2\mu + 2\sigma^2} - e^{2\mu + \sigma^2}$$

- Can be shown by advanced methods (?)

pdf, Mean and Variance of Lognormal

- Recall “Derived Distributions”

$$Y = e^X, X \sim N(\mu, \sigma^2)$$

$$F_Y(y) = \mathbf{P}(Y \leq y) = \mathbf{P}(e^X \leq y) = \mathbf{P}(x \leq \log y) = F_X(\log y)$$

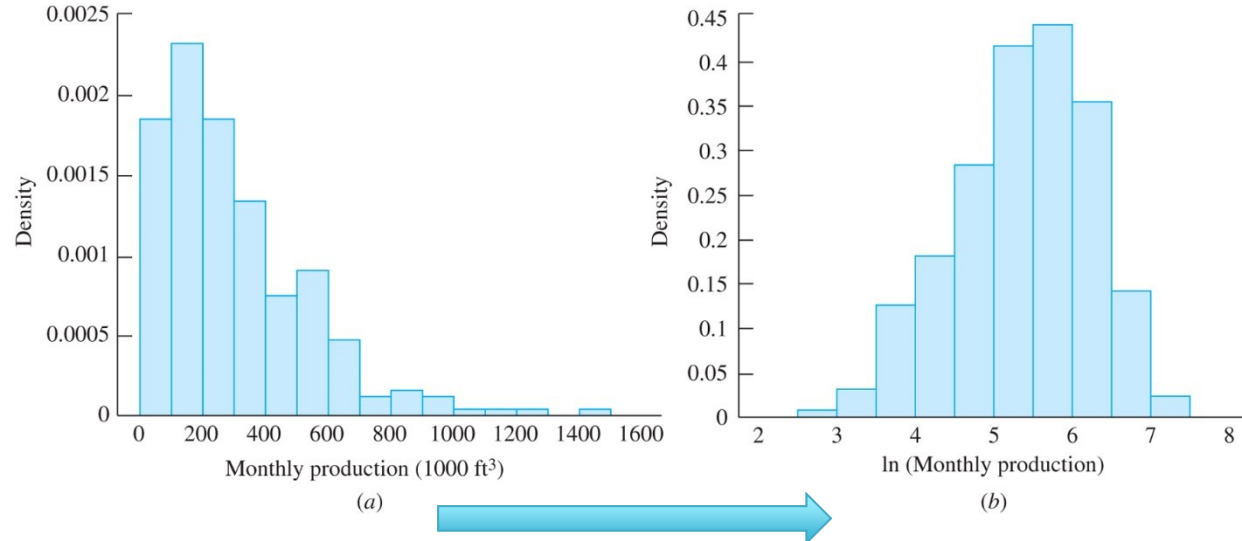
$\Rightarrow$

$$\begin{aligned} f_Y(y) &= \frac{dF_Y(y)}{dy} = \frac{dF_X(\log y)}{d \log y} \frac{d \log y}{dy} \\ &= f_X(\log y) \cdot \frac{1}{y} \\ &= \frac{1}{y\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2\sigma^2}(\log y - \mu)^2\right] \end{aligned}$$

Test of “Lognormality”

- Transform the data by taking the natural logarithm (or any logarithm) of each value
- Plot the histogram of the transformed data to determine whether these logs come from a normal population

Probability Histogram



taking the natural logarithm on the
values of the data

The Exponential Distribution

- The exponential distribution is a continuous distribution that is sometimes used to model the time that elapses before an event occurs
 - Such a time is often called a waiting time
- The probability density of the exponential distribution involves a parameter, which is a positive constant λ whose value determines the density function's location and shape
- We write $X \sim \text{Exp}(\lambda)$

pdf, cdf, Mean and Variance of Exponential

- The pdf of an exponential r.v. is

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

- The cdf of an exponential r.v. is

$$F_X(x) = \begin{cases} 0, & x \leq 0 \\ 1 - e^{-\lambda x}, & x > 0 \end{cases}$$

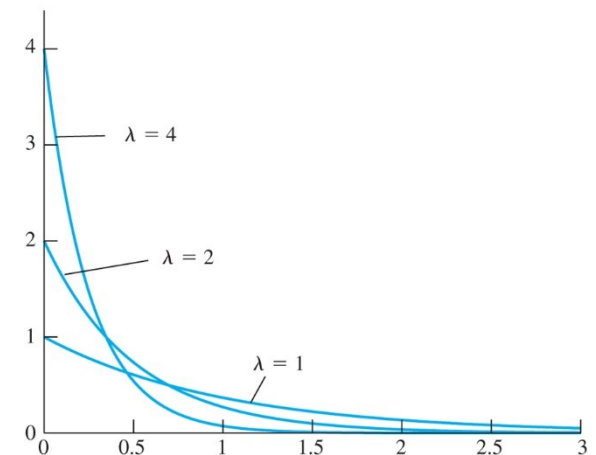
- The mean of an exponential r.v. is

$$\mu_X = 1 / \lambda$$

- The variance of an exponential r.v. is

$$\sigma_X^2 = 1 / \lambda^2$$

Probability Density Function



The Proof

- Mean and Variance of the **Exponential** Random Variable X

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & \text{if } x \geq 0, \\ 0, & \text{otherwise,} \end{cases}$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\mathbf{E}[X] = \int_0^{\infty} x \cdot f_X(x) dx = \int_0^{\infty} x \cdot \lambda e^{-\lambda x} dx$$

$$= -xe^{-\lambda x} \Big|_0^{\infty} + \int_0^{\infty} e^{-\lambda x} dx$$

$$= 0 - \frac{1}{\lambda} e^{-\lambda x} \Big|_0^{\infty} = \frac{1}{\lambda}$$

Integration by parts

$$\left(\because \frac{d(-xe^{-\lambda x})}{dx} = \lambda xe^{-\lambda x} - e^{-\lambda x} \right)$$

$$\mathbf{E}[X^2] = \int_0^{\infty} x^2 \cdot \lambda e^{-\lambda x} dx$$

$$= -x^2 e^{-\lambda x} \Big|_0^{\infty} + \int_0^{\infty} 2x e^{-\lambda x} dx$$

$$= 0 + \frac{1}{\lambda} \left(\int_0^{\infty} 2x \cdot \lambda e^{-\lambda x} dx \right) = \frac{2}{\lambda} \mathbf{E}[X] = \frac{2}{\lambda^2}$$

$$\left(\because \frac{d(-x^2 e^{-\lambda x})}{dx} = x^2 \lambda e^{-\lambda x} - 2x e^{-\lambda x} \right)$$

$$\therefore \text{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2 = \frac{1}{\lambda^2}$$

The Exponential Distribution and the Poisson Process

- If events follow a Poisson process with rate parameter λ , and if T represents the waiting time from any starting point until the next event, then $T \sim \text{Exp}(\lambda)$

Example

- Example 4.58: A radioactive mass emits particles according to a Poisson process at a mean rate of 15 particles per minute. At some point, a clock is started.
 - What is the probability that more than 5 seconds will elapse before the next emission?

The mean rate of emission is 0.25 per second, so the rate parameter is $\lambda = 0.25$, and $T \sim \text{Exp}(0.25)$.

$$\begin{aligned}\mathbf{P}(T > 5) &= 1 - \mathbf{P}(T \leq 5) \\ &= 1 - \left(1 - e^{-0.25(5)}\right) \\ &= e^{-1.25} \\ &= 0.2865\end{aligned}$$

- What is the mean waiting time until the next particle is emitted?
The mean waiting time is $\mu_T = \frac{1}{0.25} = 4$ seconds.

Example

- Suppose that a system contains a certain type of component whose time, in years, to failure is given by T . The random variable is T modeled nicely by the exponential distribution with mean time to failure equal to 5. If 5 of these components are installed in different systems, what is the probability that at least 2 are still functioning at the end of 8 years?
(假設某系統包含某種類型的元件，該類型元件在故障前使用的時間 T 以年作為計算單位，該連續隨機變數 T 為指數函數，該類型元件平均故障時間為 5 年，如果這些元件有 5 個分別安裝在不同的系統中，在 8 年結束時至少還有 2 個仍在運行的機率是多少?)

$$\mathbf{P}(X \geq a) = \int_a^{\infty} \lambda e^{-\lambda x} dx = -e^{-\lambda x} \Big|_a^{\infty} = 0 - (-e^{-\lambda a}) = e^{-\lambda a}$$

– Solution:

- $\lambda = 1/\mu_t \rightarrow \lambda = 1/5$

$$\mathbf{P}(T > 8) = \int_8^{\infty} \lambda e^{\frac{-t}{5}} dt = -e^{\frac{-t}{5}} \Big|_8^{\infty} = 0 - \left(-e^{\frac{-8}{5}}\right) = e^{\frac{-8}{5}} \approx 0.2$$

$$\mathbf{P}(X \geq 2) = \sum_{x=2}^5 \binom{n}{x} 0.2^x (1-0.2)^{5-x} = 1 - \sum_{x=0}^1 \binom{n}{x} 0.2^x (1-0.2)^{5-x} = 1 - 0.7373 = 0.2627$$

- the **average #** of outcomes **per unit time** (arrival rate)
- Also could be per unit length, area, or volume, rather than time

Lack of memory Property for Exponential

- The exponential distribution has a property known as the lack of memory property: If $T \sim \text{Exp}(\lambda)$, and t and s are positive numbers, then

$$\mathbf{P}(T > t + s \mid T > s) = \mathbf{P}(T > t)$$

$$\begin{aligned}\mathbf{P}(T > t + s \mid T > s) &= \frac{\mathbf{P}((T > t + s) \cap (T > s))}{\mathbf{P}(T > s)} \\ &= \frac{\mathbf{P}(T > t + s)}{\mathbf{P}(T > s)} = \frac{1 - F_T(t + s)}{1 - F_T(s)} \\ &= \frac{e^{-\lambda(t+s)}}{e^{-\lambda s}} = e^{-\lambda t} = 1 - F_T(t) \\ &= \mathbf{P}(T > t)\end{aligned}$$

Example

- Example 4.59: The lifetime of a particular integrated circuit has an exponential distribution with mean 2 years.
 - Find the probability that the circuit lasts longer than three years?

$$\mu_T = \frac{1}{0.5} = 2, \lambda = 0.5$$

$$\begin{aligned} T \sim \text{Exp}(0.5). \quad \mathbf{P}(T > 3) &= 1 - \mathbf{P}(T \leq 3) \\ &= 1 - (1 - e^{-0.5(3)}) \\ &= e^{-1.5} \\ &= 0.223 \end{aligned}$$

- Example 4.60: Assume the circuit is now four years old and is still functioning.
 - Find the probability that it functions for more than three additional years. Compare this probability with the probability that a new circuit functions for more than three years, which calculated in above example.

Solution to Example 4.60

– Solution:

A new circuit

$$\mathbf{P}(T > 3) = 0.223$$

$$\begin{aligned}\mathbf{P}(T > 4 + 3 \mid T > 4) &= \frac{\mathbf{P}((T > 7) \cap (T > 4))}{\mathbf{P}(T > 4)} \\ &= \frac{\mathbf{P}(T > 7)}{\mathbf{P}(T > 4)} = \frac{1 - F_T(7)}{1 - F_T(4)} \\ &= \frac{e^{-0.5(7)}}{e^{-0.5(4)}} = e^{-0.5(3)} = e^{-1.5} = 0.223\end{aligned}$$

- The probability that a 4-year-old circuit lasts 3 additional years is exactly the same as the probability that a new circuit lasts 3 years.

Estimating the Parameter of Exponential

- If $X_1, \dots, X_n$ are a random sample from $\text{Exp}(\lambda)$, then the parameter λ is estimated with $\hat{\lambda} = 1 / \bar{X}$. This estimator is **biased**. This bias is approximately equal to λ / n . The uncertainty in $\hat{\lambda}$ is estimated with

$$\sigma_{\hat{\lambda}} = \frac{1}{\bar{X} \sqrt{n}}$$

- This uncertainty estimate is reasonably good when the sample size n is more than 20

The Gamma Distribution (1/2)

- Let's consider the gamma function
 - For $r > 0$, the gamma function is defined by

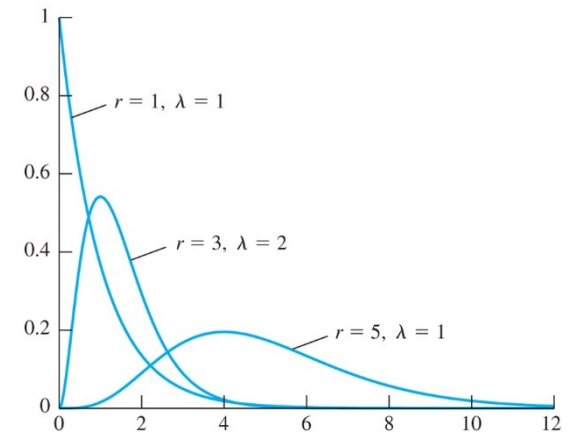
$$\Gamma(r) = \int_0^{\infty} t^{r-1} e^{-t} dt$$

- The gamma function has the following properties:
 - If r is any integer, then $\Gamma(r) = (r-1)!$
 - For any r , $\Gamma(r+1) = r\Gamma(r)$
 - $\Gamma(1/2) = \sqrt{\pi}$

The Gamma Distribution (2/2)

- The pdf of the **gamma distribution** with parameters $r > 0$ and $\lambda > 0$ is

$$f_X(x) = \begin{cases} \frac{\lambda x^{r-1} e^{-\lambda x}}{\Gamma(r)}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$



- The mean and variance of Gamma distribution are given by
 - $\mu_X = r / \lambda$ and $\sigma_X^2 = r / \lambda^2$, respectively
- If $X_1, \dots, X_r$ are independent random variables, each distributed as $\text{Exp}(\lambda)$, then the sum $X_1 + \dots + X_r$ is distributed as a gamma random variable with parameters r and λ , denoted as $\Gamma(r, \lambda)$

Example

- Example 4.65: Assume that arrival times at a drive-through window follow a Poisson process with mean $\lambda = 0.2$ arrivals per minute. Let T be the waiting time until the third arrival.
 - Find the mean and variance of T . Find $\mathbf{P}(T \leq 20)$.

$$\mu_T = \frac{r}{\lambda} = \frac{3}{0.2} = 15 \qquad \sigma_T^2 = \frac{r}{\lambda^2} = \frac{3}{0.2^2} = 75$$

$T \leq 20$ means that the third event occurs within 20 minutes. This is the same as saying that the number of events that occurs within 20 minute is greater or equal to 3.

$$\mathbf{P}(T \leq 20) = \mathbf{P}(X \geq 3)$$

Now, the mean of X is $(20)(0.2) = 4$, and X has a Poisson distribution, so $X \sim \text{Poisson}(4)$.

$$\begin{aligned} \mathbf{P}(T \leq 20) &= \mathbf{P}(X \geq 3) \\ &= 1 - \mathbf{P}(X \leq 2) \\ &= 1 - [\mathbf{P}(X = 0) + \mathbf{P}(X = 1) + \mathbf{P}(X = 2)] \\ &= 1 - \left(e^{-4} \frac{4^0}{0!} + e^{-4} \frac{4^1}{1!} + e^{-4} \frac{4^2}{2!} \right) \\ &= 1 - (e^{-4} + 4e^{-4} + 8e^{-4}) \\ &= 0.7619 \end{aligned}$$

The Weibull Distribution (1/2)

- The **Weibull distribution** is a continuous random variable that is used in a variety of situations
- A common application of the Weibull distribution is to model the lifetimes of components
- The Weibull probability density function has two parameters, both positive constants, that determine the location and shape. We denote these parameters α and β
- If $\alpha = 1$, the Weibull distribution is the same as the exponential distribution with parameter $\lambda = \beta$

The Weibull Distribution (2/2)

- The pdf of the **Weibull distribution** is

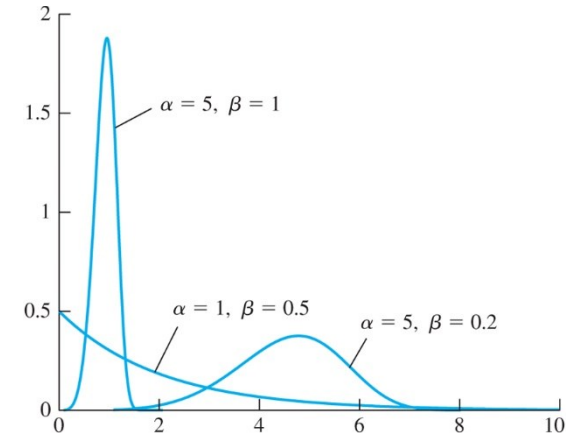
$$f_X(x) = \begin{cases} \alpha \beta^\alpha x^{\alpha-1} e^{-(\beta x)^\alpha}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

- The mean of the Weibull is

$$\mu_X = \frac{1}{\beta} \Gamma\left(1 + \frac{1}{\alpha}\right)$$

- The variance of the Weibull is

$$\sigma_X^2 = \frac{1}{\beta^2} \left\{ \Gamma\left(1 + \frac{2}{\alpha}\right) - \left[\Gamma\left(1 + \frac{1}{\alpha}\right) \right]^2 \right\}$$



Some Principles of Point Estimation

- We collect data for the purpose of estimating some numerical characteristic of the population from which they come.
- A quantity calculated from the data is called a statistic, and a statistic that is used to estimate an unknown constant, or parameter, is called **a point estimator**. Once the data has been collected, we call it **a point estimate**.
- Let $X \sim \text{Bin}(10, p)$ where p is unknown, and we observed $X = 3$, then the number $\hat{p} = 3/10$ is **a point estimate** of the unknown parameter p .

Questions of Interest

- Given a point estimator, how do we determine how good it is?
- What methods can be used to construct good point estimators?

Notation: θ is used to denote an unknown parameter, and $\hat{\theta}$ to denote an estimator of θ .

Measuring the Goodness of an Estimator

- The accuracy of an estimator is measured by its bias, and the precision is measured by its standard deviation, or uncertainty.
- To measure the overall goodness of an estimator, we used the **mean squared error (MSE)** which combines both bias and uncertainty.

Mean Squared Error

- Let θ be a parameter, and $\hat{\theta}$ an estimator of θ . The mean squared error (MSE) of $\hat{\theta}$ is

The bias of the estimator $\hat{\theta}$

$$\text{MSE}_{\hat{\theta}} = \left(\mu_{\hat{\theta}} - \theta \right)^2 + \sigma_{\hat{\theta}}^2$$

An equivalent expression for the MSE is

$$\text{MSE}_{\hat{\theta}} = \mu_{(\hat{\theta} - \theta)^2}$$

Example

- Example 4.67: Let $X \sim \text{Bin}(n, p)$ where p is unknown. Find the MSE of the estimator $\hat{p} = X/n$.

$$\text{MSE}_{\hat{\theta}} = (\mu_{\hat{\theta}} - \theta)^2 + \sigma_{\hat{\theta}}^2$$

– Solution:

We compute the bias and variance of $\hat{p}$ using above equation. As shown in Section 4.2 (Binomial distribution), the bias of $\hat{p}$ is 0, and the variance is $p(1-p)/n$. Therefore the MSE is $0 + p(1-p)/n$.

The Method of Maximum Likelihood

- The idea is to estimate a parameter with the value that makes the observed data most likely.
- When a probability mass function (pmf) or probability density function (pdf) is considered to be a function of the parameters, it is called a **likelihood function**.
- The **maximum likelihood estimate** is the value of the estimators that when substituted in for the parameters maximizes the likelihood function.

Desirable Properties

- Maximum likelihood is the most commonly used method of estimation. The main reason for this is that in most cases that arise in practice, MLEs have two very desirable properties,
 1. In most cases, as the sample size n increases, the bias of the MLE converges to 0.
 2. In most cases, as the sample size n increases, the variance of the MLE converges to a theoretical minimum.

Let $X \sim \text{Bin}(20, p)$ where p is unknown. Suppose we observe the value $X = 7$. The pmf is

$$f(7; p) = \frac{20!}{7!13!} p^7 (1-p)^{13}$$

$$\ln f(7; p) = \ln 20! - \ln 7! - \ln 13! + 7 \ln p + 13 \ln(1-p)$$

$$\frac{d}{dp} \ln f(7; p) = \frac{7}{p} - \frac{13}{1-p} = 0 \quad \Rightarrow \quad \hat{p} = \frac{7}{20}$$

Note: the quantity that maximizes the logarithm of a function is always the same quantity that maximize the function itself.

Probability (Quantile-Quantile) Plots for Finding a Distribution

- Scientists and engineers often work with data that can be thought of as a random sample from some population
 - In many cases, it is important to determine the probability distribution that approximately describes the population
- More often than not, the only way to determine an appropriate distribution is to examine the sample to find a sample distribution that fits

Finding a Distribution (1/4)

- Probability plots are a good way to determine an appropriate distribution
- Here is the idea: Suppose we have a random sample $X_1, \dots, X_n$
 - We first arrange the data in ascending order
 - Then assign increasing, evenly spaced values between 0 and 1 to each X_i
 - There are several acceptable ways to this; the simplest is to assign the value $(i - 0.5)/n$ to X_i
- The distribution that we are comparing the X 's to should have a mean and variance that match the sample mean and variance
 - We want to plot $(X_i, F(X_i))$, if this plot resembles the cdf of the distribution that we are interested in, then we conclude that that is the distribution the data came from

Finding a Distribution (2/4)

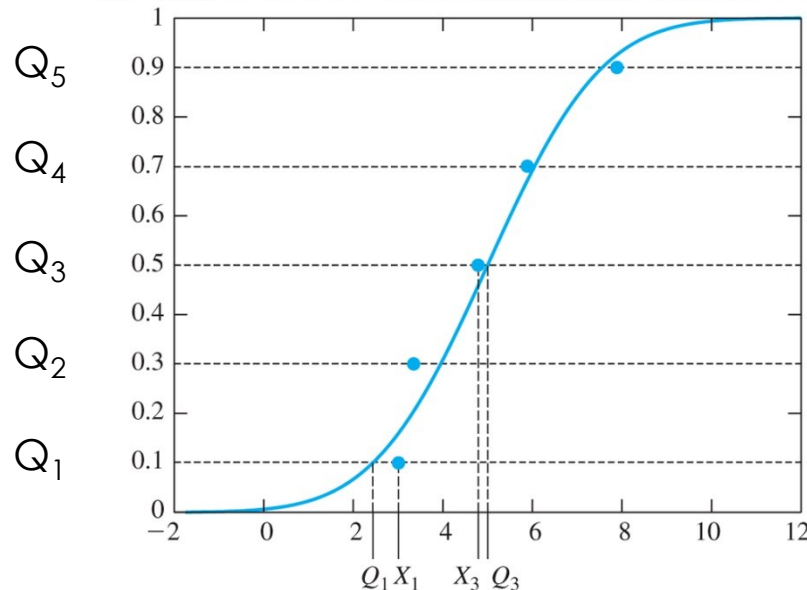
- Example: Given a sample X_i 's arranged in increasing order

3.01, 3.35, 4.79, 5.96, 7.89

i	X_i	$(i - 0.5)/5$
1	3.01	0.1
2	3.35	0.3
3	4.79	0.5
4	5.96	0.7
5	7.89	0.9

sample mean $\bar{X} = 5.00$

sample standard deviation $s = 2.00$



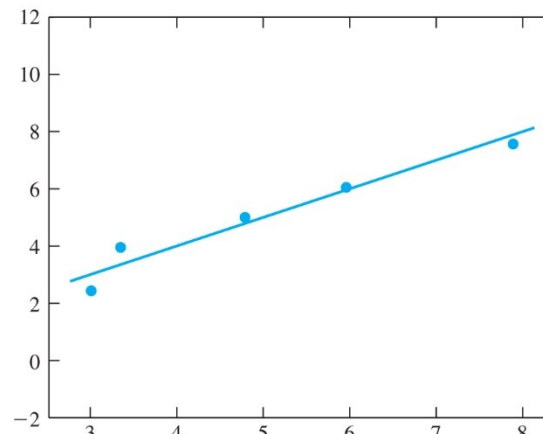
The curve is the cdf of $N(5, 2^2)$.
If the sample point X_i 's came from the distribution, they are likely to lie close to the curve.

Finding a Distribution (3/4)

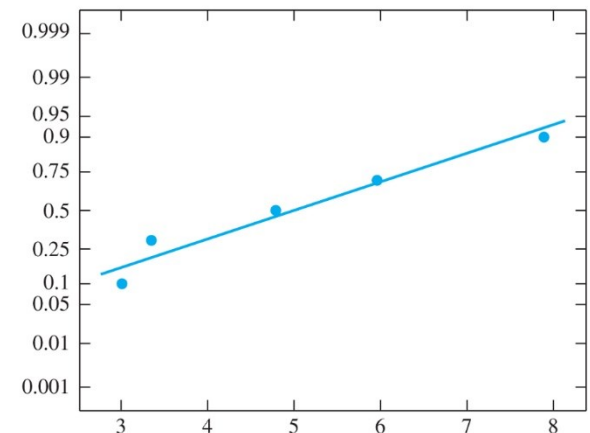
- When you use a software package, then it takes the $(i - 0.5)/n$ assigned to each X_i and calculates the quantile (Q_i) corresponding to that number from the distribution of interest. Then it plots each (X_i, Q_i) , or (Empirical Quantile, Quantile)
 - E.g., for the previous example (normal probability plot)

i	X_i	Q_i
1	3.01	2.44
2	3.35	3.95
3	4.79	5.00
4	5.96	6.05
5	7.89	7.56

Q_i



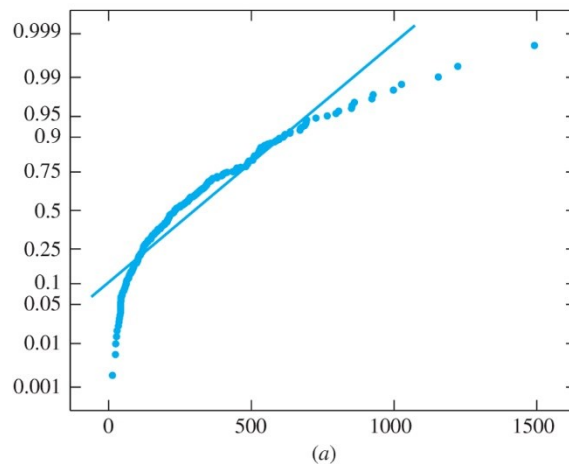
Percentile



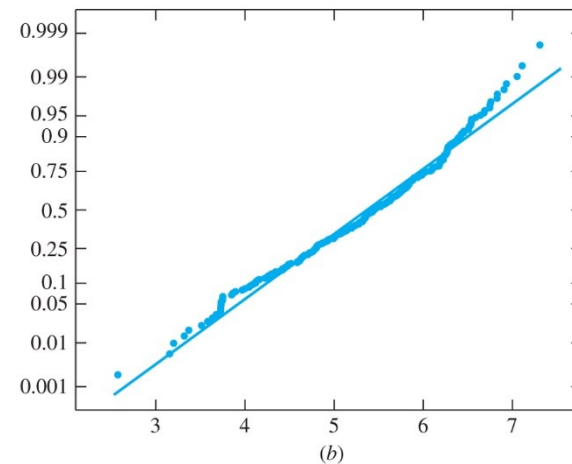
- If this plot is a reasonably straight line then you may conclude that the sample came from the distribution that we used to find quantiles

Finding a Distribution (4/4)

- A good rule of thumb is to require at least 30 points before relying on a probability plot
 - E.g., a plot of the monthly productions of 255 gas wells



monthly productions



natural logs of monthly productions

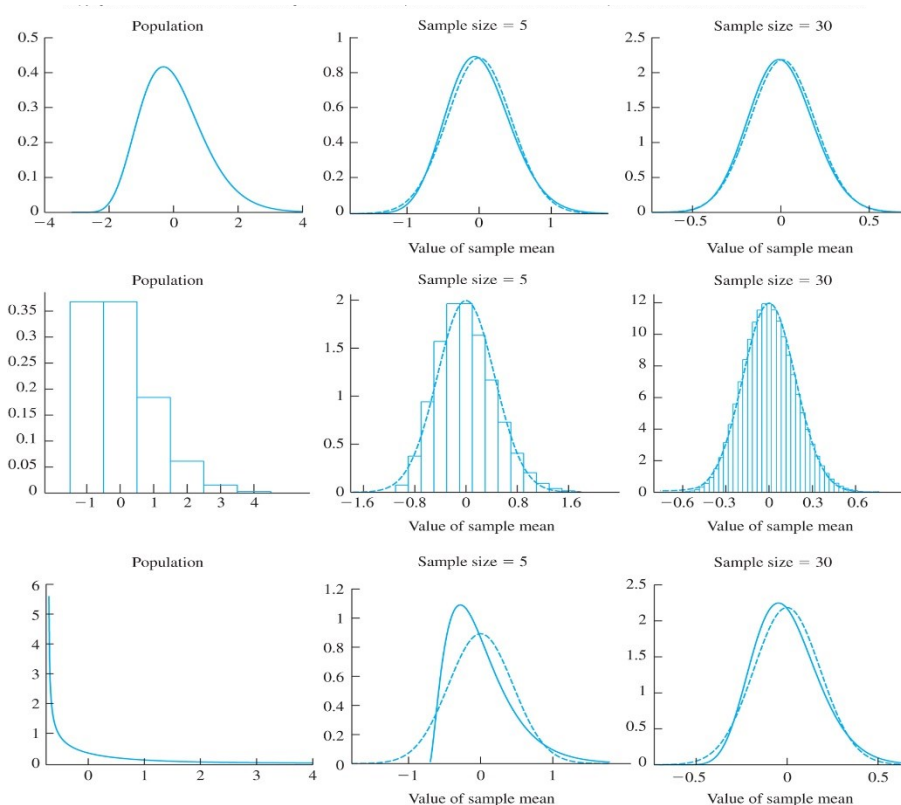
- The monthly productions follow a lognormal distribution!

The Central Limit Theorem (1/3)

- The Central Limit Theorem
 - Let $X_1, \dots, X_n$ be a random sample from a population with mean μ and variance σ^2 (n is large enough)
 - Let $\bar{X} = \frac{X_1 + \dots + X_n}{n}$ be the sample mean
 - Let $S_n = X_1 + \dots + X_n$ be the sum of the sample observations. Then if n is sufficiently large,
 - $\bar{X} = N\left(\mu, \frac{\sigma^2}{n}\right)$ sample mean is approximately normal !
 - And $S_n \sim N(n\mu, n\sigma^2)$ approximately

The Central Limit Theorem (2/3)

- Example



Solid lines: real distribution of sample mean
Dashed lines: normal distributions

- Rule of Thumb

- For most populations, if the sample size is greater than 30, the Central Limit Theorem approximation is good

The Central Limit Theorem (3/3)

- Example 4.70: Let X denotes the flaws in an 1 in. length of copper wire, and its corresponding pmf, mean and variance are

x	$P(X = x)$
0	0.48
1	0.39
2	0.12
3	0.01

$$\mu = 0.66$$

$$\sigma^2 = 0.5244$$

- One hundred wires are sampled from this population. What is the probability that the average number of flaws per wire in this sample is less than 0.5 ?

$\Rightarrow$ Following the central limit theorem, we known that

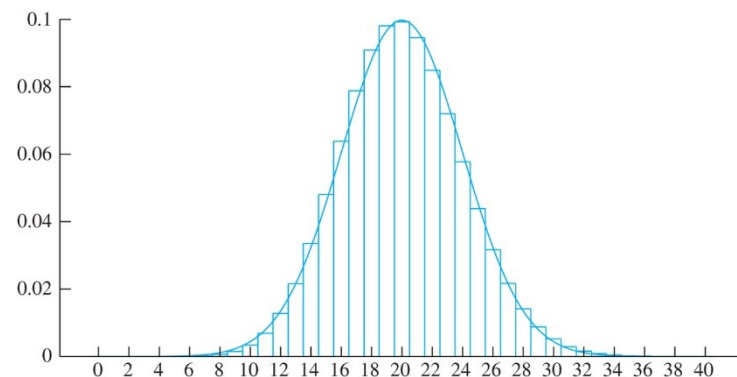
the sample mean $\bar{X} \sim N(0.66, 0.005244)$

The z - score of $\bar{X} = 0.5$

$$z = \frac{0.5 - 0.66}{\sqrt{0.005244}} = -2.21 \quad \therefore \mathbf{P}(\bar{X} < 0.5) = \mathbf{P}(Z < -2.21) = 0.0136$$

Normal Approximation to the Binomial

- $X \sim \text{Bin}(n, p)$ and if $np > 10$, and $n(1-p) > 10$, then
 - $X \sim N(np, np(1-p))$ approximately
 - And $\hat{p} = N\left(p, \frac{p(1-p)}{n}\right)$ approximately



Bin(100, 0.2) approximated by $N(20, 16)$

Recall that $X \sim \text{Bin}(n, p)$, then can X be represented as

$$X = Y_1 + Y_2 + \cdots + Y_n$$

where $Y_1, Y_2, \dots, Y_n$ is a sample from $\text{Bernoulli}(p)$

$\Rightarrow$ Following the central limit theorem, if n is large enough then

$$\hat{p} = \frac{X}{n} = \frac{Y_1 + Y_2 + \cdots + Y_n}{n} \text{ be approximated by } N\left(p, \frac{p(1-p)}{n}\right)$$

and

X can be approximated by $N(np, np(1-p))$

Normal Approximation to the Poisson

- Normal Approximation to the Poisson: If $X \sim \text{Poisson}(\lambda)$, where $\lambda > 10$, then $X \sim N(\lambda, \lambda)$
 - The Poisson can be first approximated by Binomial and then by Normal

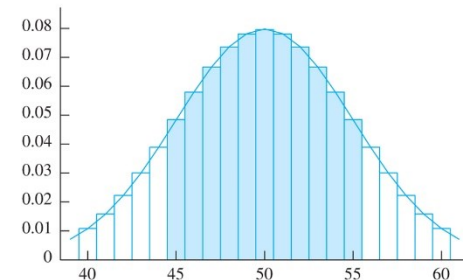
Note that variance of binomial

$$\sigma^2 = np(1-p) = \lambda(1-p) \approx \lambda \quad (\text{if } p \ll 1)$$

Continuity Correction

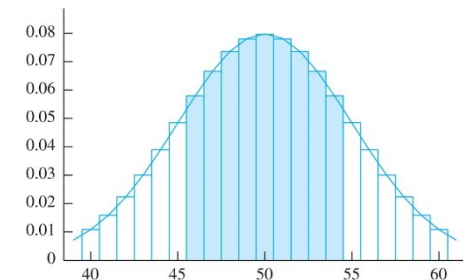
- The binomial distribution is discrete, while the normal distribution is continuous
- The continuity correction is an adjustment, made when approximating a discrete distribution with a continuous one, that can improve the accuracy of the approximation
 - If you want to **include** the endpoints in your probability calculation, then extend each endpoint by 0.5.
Then proceed with the calculation

e.g., $P(45 \leq X \leq 55)$



- If you want **exclude** the endpoints in your probability calculation, then include 0.5 less from each endpoint in the calculation

e.g., $P(45 < X < 55)$



Summary

- We considered various discrete distributions: Bernoulli, Binomial, Poisson, Hypergeometric, Geometric, Negative Binomial, and Multinomial
- Then we looked at some continuous distributions: Normal, Exponential, Gamma, and Weibull
- We learned about the Central Limit Theorem
- We discussed Normal approximations to the Binomial and Poisson distributions